

BIGQUERY CASE STUDY

LOAN PORTFOLIO RISK ANALYTICS

ADITI PAITANDY | Data Analytics Portfolio

THE CHALLENGE

The Business Problem

Balancing capital preservation with revenue growth through predictive SQL modeling and behavioral segmentation.

Executive Summary



Optimize Revenue

Identified high-performing urban clusters with 87% approval rates to maximize loan yield and market share.



Mitigate Default

Quantified the impact of credit history on default risk, identifying a 66% risk premium for non-history applicants.



Segment Growth

Developing data-backed automated approval paths for stable, high-deposit customer segments.

Data Integration Strategy

Relational Modeling

Synthesized four disparate datasets in Google BigQuery to create a 360-degree risk profile for 367 unique applicants.

```
LEFT JOIN `customers` c  
ON c.Loan_ID = l.Loan_ID
```

Leveraged cloud SQL to handle multi-table joins, ensuring 100% data integrity and consistency across demographics and balances.



Creating data integration diagram

Approval Benchmarks

367

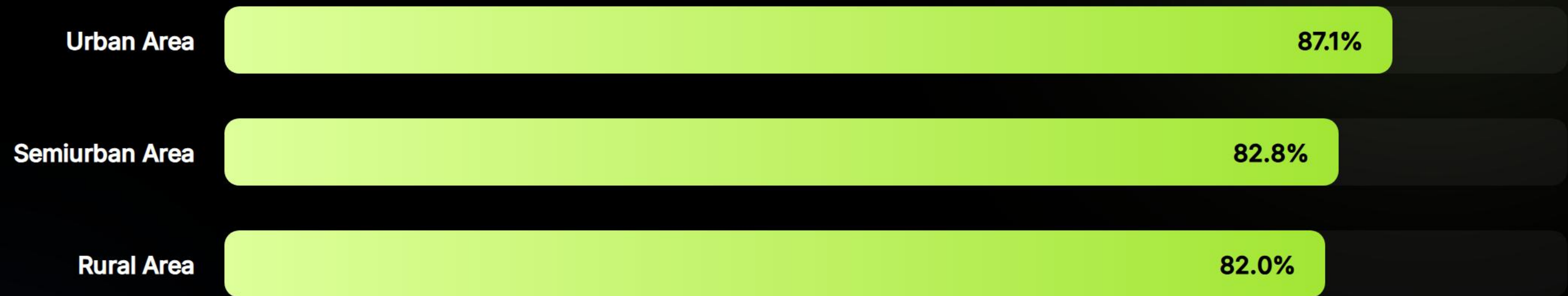
TOTAL APPLICATIONS

84.2%

BASELINE APPROVAL

Baseline metrics establish the ground truth for portfolio risk, allowing for granular variance analysis across demographics and credit scores.

Regional Risk Disparity



Urban centers emerge as the primary profitability engine, requiring specialized marketing and risk-tiering focus.

Demographic Approval Trends

Male Approval Rate

85.2%

248 Approved out of 291 Applications. High stability observed in Urban segments.

Female Approval Rate

80.3%

61 Approved out of 76 Applications. Higher concentration of rural applicants noted.

The Credit History Gatekeeper

CREDIT HISTORY STATUS	APPLICANT COUNT	APPROVED COUNT	SUCCESS RATE %
1 (Healthy History)	279	279	100.0%
0 (Negative History)	88	30	34.1%

Observation: History = 1 is the primary non-negotiable filter. Rejection recovery should focus on History = 0 applicants with high stability scores.

The Income Stability Paradox



Stability Over Volume

Data suggests that raw income is a noise factor. High earners with volatile expenditure profiles represent higher default risks than mid-earners with high deposit stability.

- ✓ Focus on "Deposit Velocity" metrics.
- ✓ Prioritize Urban stability over Rural volume.

LET'S CONNECT

Bridging SQL Excellence with Banking Strategy

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